

Diagnostic accuracy of single baseline measurement of Elecsys Troponin T high-sensitive assay for diagnosis of acute myocardial infarction in emergency department: systematic review and meta-analysis

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Evidence abstract

OBJECTIVE To obtain summary estimates of the accuracy of a single baseline measurement of the Elecsys Troponin T high-sensitive assay (Roche Diagnostics) for the diagnosis of acute myocardial infarction in patients presenting to the emergency department.

DESIGN Systematic review and meta-analysis of diagnostic test accuracy studies.

DATA SOURCES Medline, Embase, and other relevant electronic databases were searched for papers published between January 2006 and December 2013.

STUDY SELECTION Studies were included if they evaluated the diagnostic accuracy of a single baseline measurement of Elecsys Troponin T high-sensitive assay for the diagnosis of acute myocardial infarction in patients presenting to the emergency department with suspected acute coronary syndrome.

STUDY APPRAISAL AND DATA SYNTHESIS The first author screened all titles and abstracts identified through the searches and selected all potentially relevant papers.

The screening of the full texts, the data extraction, and the methodological quality assessment, using the adapted QUADAS-2 tool, were conducted independently by two reviewers with disagreements being resolved through discussion or arbitration.

If appropriate, meta-analysis was conducted using the hierarchical bivariate model.

RESULTS Twenty three studies reported the performance of the evaluated assay at presentation.

The results for 14 ng/L and 3-5 ng/L cut-off values were pooled separately.

At 14 ng/L (20 papers), the summary sensitivity was 89.5% (95% confidence interval 86.3% to 92.1%) and the summary specificity was 77.1% (68.7% to 83.7%).

At 3-5 ng/L (six papers), the summary sensitivity was 97.4% (94.9% to 98.7%) and the summary specificity was 42.4% (31.2% to 54.5%).

This means that if 21 of 100 consecutive patients have the target condition (21%, the median prevalence across the studies), 2 (95% confidence interval 2 to 3) of 21 patients with acute myocardial infarction will be missed (false negatives) if 14 ng/L is used as a cut-off value and 18 (13 to 25) of 79 patients without acute myocardial infarction will test positive (false positives).

If the 3-5 ng/L cut-off value is used, <1 (0 to 1) patient with acute myocardial infarction will be missed and 46 (36 to 54) patients without acute myocardial infarction will test positive.

CONCLUSIONS The results indicate that a single baseline measurement of the Elecsys Troponin T high-sensitive assay could be used to rule out acute myocardial infarction if lower cut-off values such as 3 ng/L or 5 ng/L are used.

However, this method should be part of a comprehensive triage strategy and may not be appropriate for patients who present less than three hours after symptom onset.

Care must also be exercised because of the higher imprecision of the evaluated assay and the greater effect of lot-to-lot reagent variation at low troponin concentrations.

SYSTEMATIC REVIEW REGISTRATION PROSPERO registration number CRD42013003926.

Document metadata

